Forecasting daily Potomac discharge and hourly Gandak Barrage inflow

Implications for water allocation and flood warning

Abstract

River forecasts inform decisions at different time scales, from allocating the next day's water supply to anticipating a flood crest within hours. This study evaluates next-day discharge forecasts for the Potomac River in the United States and hourly inflow forecasts for Gandak Barrage in India. In each basin, forecast skill was assessed against persistence, which assumes that discharge remains at its latest observed value.

For the Potomac, daily observations from 1990 to 2025 supported a comparison of three long short-term memory (LSTM) architectures, a Transformer, and a Bayesian model averaging ensemble. Across 2,192 test days in 2020–2025, the stacked LSTM achieved the lowest root-mean-square error (111.22 m³/s, compared with 145.79 m³/s for persistence). The ensemble achieved the lowest mean absolute error (37.79 m³/s), and its 90% prediction interval contained 93.0% of observed flows. In a river-basin digital-twin simulation with hypothetical demand, allocating water from the fifth percentile of forecast supply rather than the ensemble mean reduced the proportion of days when allocations exceeded available supply from 63.5% to 1.5%, while increasing unmet demand.

For Gandak Barrage, hourly inflow records from August to October 2026 were used to fit and test ridge regression models at leads of one to 72 hours. Persistence yielded lower mean absolute error at every lead. At 24 hours, the error was 732 m³/s for persistence and 1,025 m³/s for regression. The forecast issued on 27 September illustrates the consequence for flood warning: the model predicted a maximum inflow of 12,519 m³/s over the next 72 hours, but recorded inflow reached 15,396 m³/s after only 14 hours.

The Potomac results quantify the trade-off between overallocating water and leaving demand unmet. At Gandak, the missed September crest identifies a specific question for further testing: whether upstream observations at Devghat and rainfall forecasts available at issue time can provide earlier warning than recent barrage inflow alone.